

Computational Fluid Dynamics

Prof. Parolini and Prof. Valdettaro

2/9/2024

We want to simulate the flow of fluid with kinematic viscosity $\nu = 0.01$ through the bifurcating duct sketched in Figure 1. The problem is governed by the the unsteady incompressible Navier-Stokes equations, with Dirichlet boundary conditions imposed on the inflow left boundary

$$\mathbf{u} = [(2 - y)(2 + y)/4, 0]^T, \quad \text{on } \Gamma_1,$$

and on the duct walls

$$\mathbf{u} = [0, 0]^T, \quad \text{on } \Gamma_4,$$

(as indicated in the Figure). On the top branch outlet Γ_3 , we impose a zero normal stress boundary condition

$$-p\mathbf{n} + 2\nu \left(\frac{\nabla \mathbf{u} + \nabla \mathbf{u}^T}{2} \right) \mathbf{n} = \mathbf{0},$$

while on the bottom branch outlet Γ_2 , a time-dependent non-homogeneous Neumann condition is imposed, namely

$$-p\mathbf{n} + 2\nu \left(\frac{\nabla \mathbf{u} + \nabla \mathbf{u}^T}{2} \right) \mathbf{n} = \mathbf{d},$$

with $\mathbf{d} = (\frac{1}{100} \sin(2\pi t), 0)^T$.

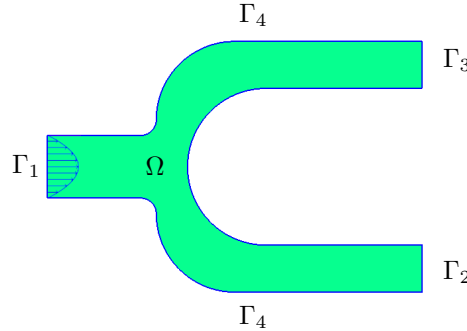


Figure 1: Computational domain

- a) Introduce a time-advancing scheme using the **non-incremental Chorin-Temam** projection method.
- b) Write the weak formulations of the problems to solve at each time iteration discussing the choice of functional spaces and the treatment of boundary conditions.
- c) Write a Firedrake notebook, based on the template `template020924.ipynb`, implementing the proposed scheme using $\mathbb{P}_2$ finite elements for velocity and $\mathbb{P}_1$ finite elements for pressure.
- d) Compute the solution, starting from rest (zero velocity and pressure) until the final time $t_f = 100$ with a time step $\Delta t = 5$. Then:
 - at each time step compute the flowrates through each of the two outlet boundaries, display the two flowrates versus time and export the plot in a `.png` file;
 - save plots of the velocity and pressure fields at times $t = 50$ and $t = 100$ (export them as `.png` files);
 - based on the results, discuss the physical behaviour of the system.